

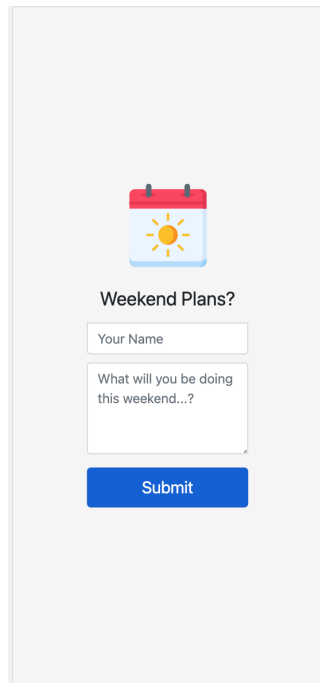
CheckMate

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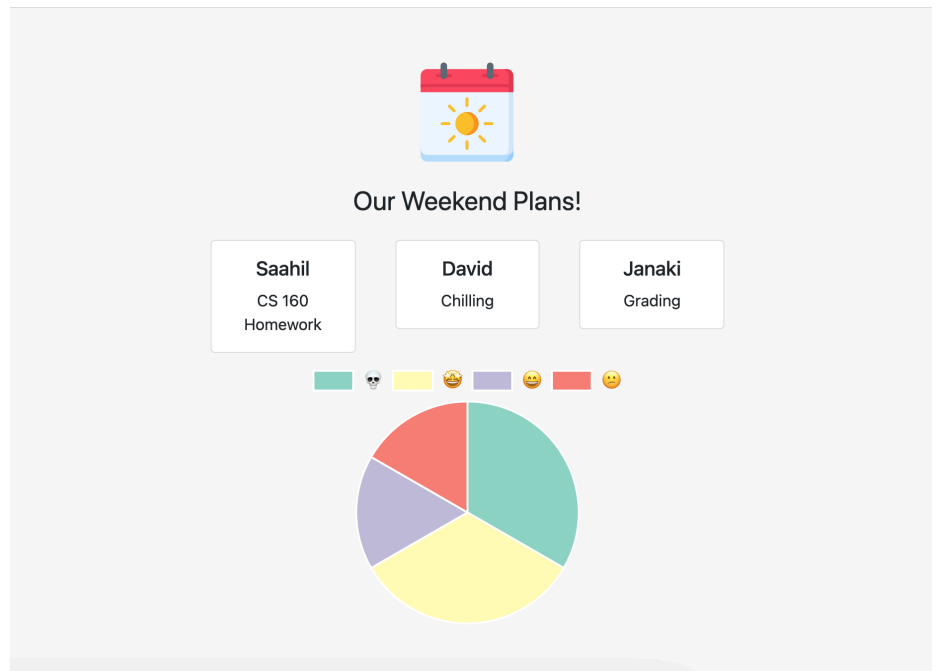
Group #11

Tuesday, July 27, 2021

CS 160, Summer '21



The image shows a mobile app interface for 'CheckMate'. At the top is a calendar icon with a sun. Below it is the title 'Weekend Plans?'. There is a text input field labeled 'Your Name' and another text input field labeled 'What will you be doing this weekend...?'. At the bottom is a blue 'Submit' button.



How we interact with coworkers has dramatically changed in the new virtual world brought on by the pandemic. Although the productivity aspect of the workplace has adapted, companies adopting new technologies like Slack or Zoom to continue meeting deadlines and delivering results, the informal side has not adapted quite as well. Getting to know your coworkers was a big part of corporate culture; it contributed to morale and mitigated feelings of burnout. This perceived lack of casual communication with coworkers was brought up repeatedly in our user interviews, and it is what we are addressing through this application.

CheckMate is a platform for coworkers to connect with each other. Users complete a daily check-in question on their mobile device at the beginning of a work meeting. As they fill out this form, they can see their results populate on the web view. This helps team managers take a temperature check of their employees while also empowering coworkers to learn more about one another and bond. Through these check-in questions, we hope to bring casual, informal communication between employees back to the workplace. *CheckMate: Check on your Mate* 😊

Application Context

Description of Target Users

Our target user is between the ages of 20 and 60. They are working professionals in a wide variety of industries, ranging from software to healthcare to finance. They work standard 9-5 corporate jobs. After a year at home, some are finally going back to the office and are looking for ways to boost their productivity while in meetings.

Description of Collaborative Context

The context of the app is a group of professionals who are looking to boost their productivity, to be more efficient during meetings, share resources, and facilitate communication in the workplace. The collaboration encourages collaboration across departments and more interaction and input into executive decision making.

Narrative Scenarios

Post-it Mood Board

Alex and his marketing team wants to rebrand a product for his company. He wants to use a virtual platform that is professional and visually appealing, so he can present the company with his ideas. Alex asks other departments for input and feedback, but without a visual representation, it is hard to encapsulate their goals and vision while rushing through a 5-minute elevator pitch. He decides to download a post-it note mood board app to organize his ideas. Alex uses it during his team's brainstorming sessions to record and envision ideas and iterations. He then uses email and slack to share his visual platform to the entire workplace so that they can be updated on the rebrand. His co-workers directly comment and put up post-it notes on the board to provide feedback when they have free time.

Task manager

Ashley and her team of six software engineers are in their bi-weekly sprint planning meeting on Monday morning in a company boardroom. The engineering team follows agile practices, so prior to the meeting, each engineer had written user stories describing the tasks and acceptance criteria for the work they want to complete this sprint. At the beginning of the meeting, each engineer uses their own phone to go through the backlog of tasks written by everyone, leaves comments, and anonymously puts in their estimate for the number of half-days (task points) they estimate the task will take. After everyone has done their independent assessments, Ashley brings her team together and they go through each task on the big projector screen. At this point, the team reviews comments, finalizes acceptance criteria, and determines a final estimate for the task.

Daily check-in

Dexter leads a design team at a PR firm. When the pandemic began, he noticed a significant drop in morale among his team. He realized that work-from-home culture was missing the kind of informal workplace interactions that were common when they were in person. So, Dexter reserves the beginning of his team's daily all-hands meeting to do a check-in. Every team member anonymously answers an icebreaker question like "what are your weekend plans?" and the results get populated on the screen for everyone to see.

Interviews

Recruiting Message

Hi [name] 🙌! I'm in a course called "User Interface Design and Development." My partner and I are interested in learning how working professionals in a wide array of industries and roles work collaboratively in meetings and other productivity tasks. Would you be open to helping us with our study?

In this study, you'll get the opportunity to contribute to the development of a collaborative productivity tool! Participants will sign up for a 30-minute session where you'll be asked a series of questions to understand how, why, and when you collaborate with others in the workplace. We'll ask you to recount real-life examples and situations where you've worked on a problem with others or alone.

If you're interested in participating in our study, please fill out this [Google Form](#)!

Interview Guide

- Walk us through your day to day interactions.
- What is the company culture and attitude towards collaboration and sharing information?
- Are there any communication roadblocks (between departments, with executives...etc)?
- What current technologies and tools does your company utilize for communication and collaboration?
- Tell me about a time when you had to collaborate with others in a stressful situation. What was the situation, why was it stressful, what tools did you use?
- How does your team work together to decide what work needs to be done and how that work will be distributed across the team?
- What does a typical presentation look like? How do presenters solicit feedback, if at all? Tell me about a time when you were a presenter, and tell me about a time when you were a listener.
- Because of COVID, most corporate professionals had to adjust to remote work for the first time. How did this impact your ability to collaborate with your colleagues? Can you give an example?
- COVID also eliminated the ability for the kind of spontaneous physical contact that was common in office environments. The lack of this 'watercooler conversation' posed a challenge for team bonding and informal communication. Can you tell us how this impacted you? Has your team done anything to compensate for this lack? If so, what did you like or dislike about this?

Participants

#	Description	Interview Length
1	This user is an 18-year-old intern at a large healthcare organization. He is a business analyst intern working on prior authorizations for COVID-19. He works full-time remotely from home.	30 min
2	This user is a 27-year-old employee at a large e-commerce company. He has worked as a software development engineer at this company for 5 years. He is also working remotely from home.	30 min

Synthesized Findings

Strong Use of Collaboration Tools

Both participants discussed their heavy usage of collaboration tools. The first participant cited Microsoft Teams, and the second participants talked about Slack. The participants said that the existing tools that they use on a day-to-day basis serve them well, and the main pain points are associated with losing the informal communication from before the pandemic, walking over to someone else's desk to ask them a quick question instead of scheduling a 30-minute call with them.

Lack of Informal Communication

Both participants feel like their social communication with their colleagues is lacking. The first participant says that no effort has been made on his team to recreate the interpersonal communication that existed before the pandemic; there are no happy hours or coffee breaks. The second participant says that there has been some effort, a biweekly happy hour. But this is also not as good as it could be, since several participants don't turn on their cameras.

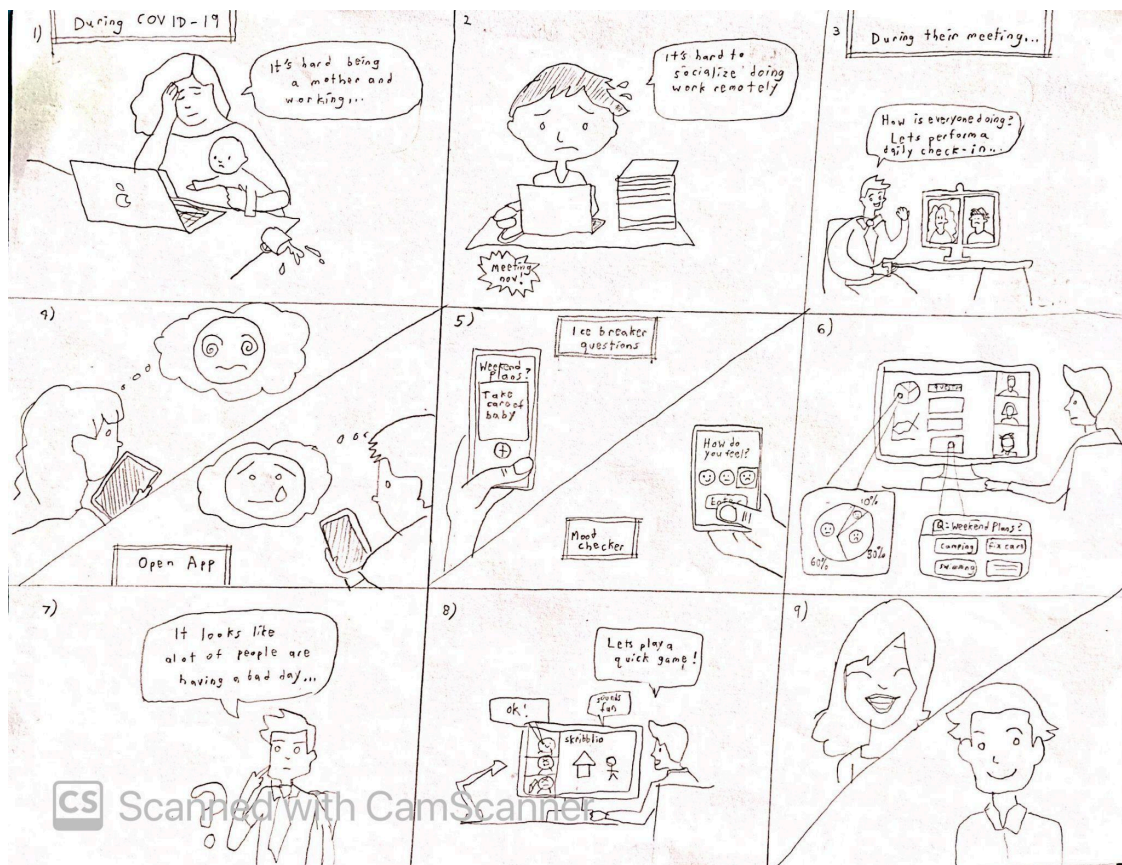
Application Design

Persona

Dexter is a university student working for a software company. His end goal is to work for a Cybersecurity company that prioritizes work life balance. When the pandemic began, he noticed a significant drop in morale and his mental health due to online classes and zoom fatigue. From his past intern experiences, he has realized that work-from-home culture was missing the kind of informal workplace interactions that were common when they were in person. This made it harder to socialize and meet new people in the workplace. So, Dexter uses the app at the beginning of his team's daily all-hands meeting to do a check-in. Every team member anonymously reports their mood, and answers an icebreaker question like "what are your weekend plans?" and the results get populated on the screen for everyone to see. Doing this makes him feel acknowledged and supported by his team and project manager.



Storyboard

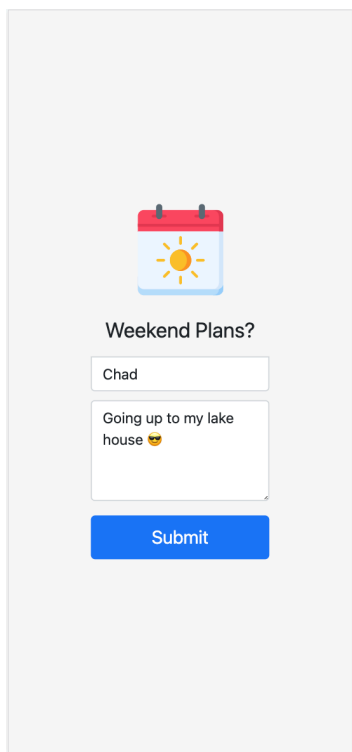


Interactive Prototype

Interface and Interactions

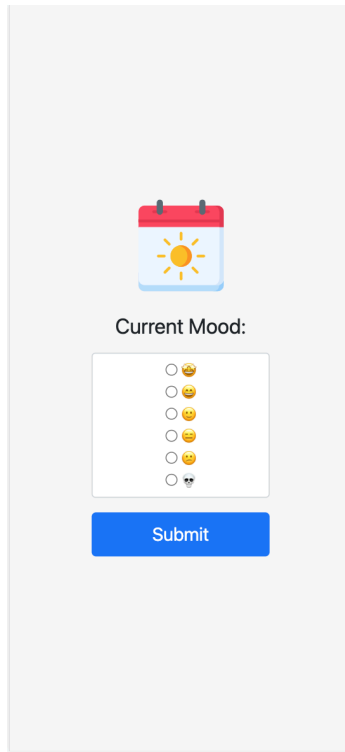
At the beginning of a daily stand-up meeting, each team member completes a check-in question on their mobile device, and results are populated on the web view. Some participants receive the question “what are your weekend plans?” and complete a text-based response. The web view displays all responses. Others are asked “How do you feel?” and select from a set of emojis. These results are aggregated into a pie chart. By controlling which type of question the user receives, and how many answer choices they have available, we test how this affects time to answer.

Screenshots

A mobile app interface for a text-based check-in. At the top is a calendar icon with a sun. Below it, the text "Weekend Plans?" is displayed. There are two text input fields: the first contains "Chad" and the second contains "Going up to my lake house 🌞". A blue "Submit" button is at the bottom.

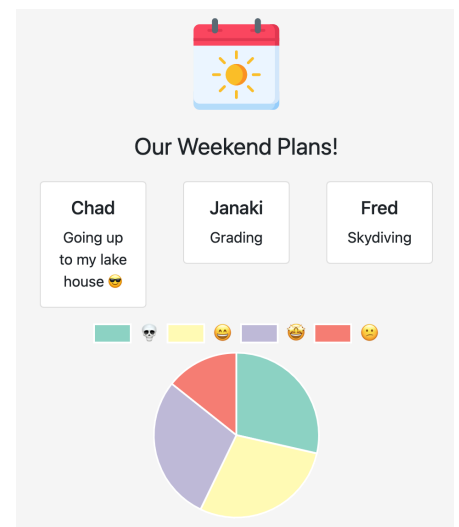
Mobile - Text

The text-based check-in question lets users describe their weekend plans.

A mobile app interface for an emoji-based check-in. At the top is a calendar icon with a sun. Below it, the text "Current Mood:" is displayed. There is a vertical list of seven emoji options: 😊, 😊, 😊, 😊, 😊, 😊, and 🤪. A blue "Submit" button is at the bottom.

Mobile - Emoji

In the emoji-based check-in question, users select how they're feeling from a list of emoji options.



Web

The web view displays aggregation of all the users' responses.

*Links***Github repository**<https://github.com/cs160-summer-2021/p4-starter-code-saahilchadha1>**Final video**<https://www.youtube.com/watch?v=83rMcrgtjU4>

Experiment

Experimental Design

Chosen Key Question	How many moods are enough, or too much to choose from?
Hypothesis	According to Hicks law, the more choices a user is presented with, the longer it will take for them to reach a decision. Therefore, we hypothesize that having 3 moods (happy, okay, sad) is enough to satisfy but not overwhelm a user.
Independent Variables	The independent variables are the number, and options of moods that a user can select from. Each user is exposed to a set of 3, 5 and 7 emoji options per screen.
Dependent Variables	The dependent variables are how long it takes for a user to select a mood, and whether he/she has enough options to sufficiently express how they feel.
Instrumentation	We need to collect the amount of time it takes for a user to select their mood by timing how fast a user does something (from the page loading to clicking the submit button). We will do this using a stopwatch application.
Survey Questions	Do you feel like there were enough options to express your mood? Which number of options did you prefer? Would you add another option? Was the variety of options overwhelming? Do you prefer emojis or words?
Within subjects or between subjects	We will perform a within subject experiment. We will randomize the order of options and tasks that the users complete. Each task will be shown on a separate screen and the user will be presented with instructions beforehand to learn the task.
Tasks	Each task consists of the participant searching to select an option that best represents their mood. One task will consist of 7 emojis to choose from, the other tasks consists of 5 options,, and 3 options.

Participants and Conditions

Name	Conditions
Nitya Krishnakumar	Text, 3-option emoji, 5-option emoji, 7-option emoji. Random order
Noah Schwartz	Text, 3-option emoji, 5-option emoji, 7-option emoji. Random order
Calvin Pham	Text, 3-option emoji, 5-option emoji, 7-option emoji. Random order
Shreya Puli	Text, 3-option emoji, 5-option emoji, 7-option emoji. Random order

Results

The tables below show the data that we collected through our usability studies. The amount of time taken to complete the check-in form was generally shorter for the 3-option variant than for the 5 and 7-option ones. The text time is also generally higher than all the multiple choice options. In the exit interview questions, all participants found that the 7-option variant was sufficient to express their mood. And, they all preferred 7 or 5 to the 3-option one, which many found insufficient. Half participants thought that 7 options are optimal before becoming overwhelming, and the other half found that any number above 5 is overwhelming. All participants unanimously preferred emojis over words to express their mood.

Name	Text Time	3-option Time	5-option Time	7-option Time
Nitya Krishnakumar	7:43	1:34	6:99	4:50
Noah Schwartz	6:32	2:78	17:56	8:56
Calvin Pham	34:23	8:15	9:43	6:80
Shreya Puli	29:20	9:03	6:49	3:09

Name	Enough options to express mood?	Which number of options did you prefer? Would you add another option?	Was the variety of options overwhelming	Do you prefer emojis or words?
Nitya Krishnakumar	3 was too little, hard to make decision	Prefer 5 bc 3 too little	Having 6 was a little overwhelming	Preferred emojis over words
Noah Schwartz	It sufficed	Preferred 7	More than 7 would be overwhelming	Emotes are easier to visually see the meaning
Calvin Pham	It was okay, but preferred more options	Preferred 7, add a neutral emoji (middle looks annoyed rather than neutral)	Having more than 5 overwhelms	Emoji
Shreya Puli	Yes. I liked the additional non-face emoji	Preferred 7. No, this was a good amount	No, more than 7 would be overwhelming	Emojis

Reflection on Hypothesis

The key question we sought to answer was: How many moods are enough, or too much to choose from? Our hypothesis was that the more choices a user is presented with, the longer it will take for them to reach a decision, as according to Hick's Law. Therefore, we hypothesized that having 3 moods (happy, okay, sad) is enough to satisfy, but not overwhelm a user. From our results, we saw that for each user, the response rate was quickest for 3 emojis, which aligns with Hick's Law: the average response time was 5:32 seconds. However, with 5 options, the response rate peaked at an average of 10:12 seconds, most likely due to having to compromise between options. With 7 options however, users had an average response rate of 5.72 seconds, similar to that of 3 options. When asked which number of options they preferred, $\frac{3}{4}$ users responded that they preferred 7 options, and all stated that more than 7 would be overwhelming. When asked if 3 options were enough to express mood, many said that it sufficed but they preferred more options. From these findings, we were able to find compelling evidence towards having 7 options as the optimal number that would be enough to suffice users, and garner a quick response rate.